

AI-Powered Student Stress Prediction and Wellness Recommendation System

“The greatest wealth is health, and mental well-being is its foundation.”

ABSTRACT

Student mental health has emerged as a significant concern in modern educational environments due to increasing academic pressure, excessive screen exposure, irregular sleep patterns, and changing lifestyles. Stress among students can negatively impact academic performance, emotional stability, physical health, and overall well-being. Therefore, the early identification of stress levels is essential for promoting a healthier and more productive learning environment.

The proposed project, 'AI-Powered Student Stress Prediction and Wellness Recommendation System,' is designed to predict the stress levels of students using Machine Learning techniques and provide personalized wellness recommendations. The system analyzes key factors influencing student well-being, including sleep hours, study hours, screen time, exercise hours, attendance percentage, social activity, and academic pressure. These inputs are processed by a trained Machine Learning model to classify students into four categories: Low Stress, Moderate Stress, High Stress, and Critical Stress.

The application is developed using Python, Flask, HTML, CSS, SQLite, and Machine Learning libraries to create an interactive and user-friendly platform. Based on the predicted stress level, the system generates personalized wellness recommendations and calculates a wellness score to help students understand their current mental wellness condition. Additionally, the system stores prediction history in a database, enabling users to review previous assessments and monitor stress patterns over time.

The proposed solution provides an intelligent, data-driven, and accessible approach to student wellness management. By combining Artificial Intelligence with mental health awareness, the system supports early stress detection, encourages healthy lifestyle habits, and promotes proactive wellness practices among students. Furthermore, the project can be enhanced in the future through cloud deployment, mobile application integration, advanced analytics, and real-time monitoring capabilities.

Keywords: Artificial Intelligence, Machine Learning, Student Wellness, Stress Prediction, Mental Health, Flask, SQLite, Wellness Recommendation System.

INTRODUCTION

"The future of a nation is shaped in its classrooms, and the well-being of students determines the strength of that future." "Healthy minds inspire innovation, productivity, and lifelong success."

Mental health plays a vital role in the academic, emotional, and social development of students. In today's competitive educational environment, students are often exposed to various stress-inducing factors such as academic workload, examination pressure, limited physical activity, excessive screen time, irregular sleep schedules, and reduced social interaction. These factors can significantly affect a student's concentration, decision-making ability, academic performance, and overall quality of life.

Stress is a natural response to challenging situations; however, prolonged or unmanaged stress can lead to serious mental and physical health issues. Many students fail to recognize the early signs of stress, which may result in decreased productivity, anxiety, depression, and other psychological concerns. Therefore, there is a growing need for intelligent systems that can assist in identifying stress levels at an early stage and provide appropriate guidance to improve student well-being.

Recent advancements in Artificial Intelligence (AI) and Machine Learning (ML) have enabled the development of predictive systems capable of analyzing behavioral and lifestyle patterns. These technologies can process multiple factors simultaneously and generate meaningful insights that support decision-making. By leveraging AI and ML, educational institutions can adopt proactive approaches toward student wellness management rather than relying solely on reactive interventions.

The proposed project, "AI-Powered Student Stress Prediction and Wellness Recommendation System," aims to utilize Machine Learning techniques to predict student stress levels based on key lifestyle and academic parameters. The system collects inputs such as sleep hours, study hours, screen time, exercise hours, attendance percentage, social activity, and academic pressure. Based on these factors, the trained Machine Learning model classifies students into four stress categories: Low Stress, Moderate Stress, High Stress, and Critical Stress.

In addition to stress prediction, the system provides personalized wellness recommendations and generates a wellness score to help students understand their mental wellness status. The application also maintains prediction history using a database, enabling users to monitor their stress patterns over time. By integrating predictive analytics with wellness guidance, the proposed system promotes early stress detection, encourages healthy lifestyle habits, and contributes to creating a supportive educational environment.

This project demonstrates how Artificial Intelligence can be applied beyond traditional automation tasks to address real-world challenges related to student mental health and well-being. It serves as a practical solution that combines technology, healthcare awareness, and educational support to enhance the overall student experience.

PROBLEM STATEMENT

"Every challenge presents an opportunity for innovation and positive change." "Understanding stress today can help build healthier and more successful students tomorrow."

Student mental health has become a growing concern in educational institutions across the world. Rapid changes in academic environments, increasing competition, examination pressure, excessive use of digital devices, irregular sleep patterns, and reduced physical activity have significantly contributed to rising stress levels among students. While moderate stress can motivate students to perform better, prolonged or excessive stress can negatively impact academic performance, emotional well-being, physical health, and overall quality of life.

One of the major challenges faced by educational institutions is the lack of effective mechanisms for the early identification of student stress. In many cases, students fail to recognize the warning signs of stress until it begins to affect their academic achievements and mental health. Traditional approaches often rely on manual observation, counseling sessions, or self-reporting methods, which may not always provide timely or accurate assessments.

Furthermore, existing student support systems generally focus on addressing stress after it has already become a serious concern rather than preventing it at an early stage. This reactive approach limits the ability of educators and students to take proactive measures for maintaining mental wellness. There is a growing need for intelligent and data-driven solutions that can continuously evaluate factors influencing student well-being and provide timely recommendations for stress management.

The absence of a simple, accessible, and automated stress prediction system creates a gap between mental health awareness and practical intervention. Students require a solution that not only predicts their stress levels accurately but also offers personalized wellness guidance to help them improve their daily habits and overall mental health.

To address these challenges, this project proposes an AI-Powered Student Stress Prediction and Wellness Recommendation System that utilizes Machine Learning techniques to analyze student lifestyle and academic factors. The system aims to predict stress levels at an early stage, categorize students based on risk levels, provide personalized wellness recommendations, and maintain historical records for continuous monitoring. By enabling proactive stress management, the proposed system contributes to improved student well-being, academic success, and a healthier educational environment.

OBJECTIVES

"Great achievements begin with clear objectives and a meaningful purpose." "Technology reaches its true potential when it serves humanity and improves lives."

The primary objective of this project is to develop an intelligent and user-friendly system that utilizes Artificial Intelligence and Machine Learning techniques to predict student stress levels and provide personalized wellness recommendations. The system is designed to support early stress detection, promote mental well-being, and encourage healthier lifestyle practices among students.

The specific objectives of the project are as follows:

1. To develop a Machine Learning-based model capable of predicting student stress levels using academic and lifestyle-related factors.
2. To analyze key parameters such as sleep hours, study hours, screen time, exercise hours, attendance percentage, social activity, and academic pressure that influence student stress.
3. To classify students into four stress categories: Low Stress, Moderate Stress, High Stress, and Critical Stress for effective stress assessment.
4. To provide personalized wellness recommendations based on the predicted stress level in order to encourage positive behavioral and lifestyle changes.
5. To calculate and display a wellness score that helps students understand their overall mental wellness condition.
6. To design and develop a user-friendly web application using Flask, HTML, and CSS for easy interaction and accessibility.
7. To maintain a prediction history database using SQLite for tracking previous assessments and monitoring stress patterns over time.
8. To promote awareness regarding student mental health and encourage proactive stress management practices.
9. To demonstrate the practical application of Artificial Intelligence in solving real-world challenges related to student wellness and educational support.
10. To create a scalable foundation that can be extended in the future with advanced analytics, mobile applications, cloud deployment, and real-time wellness monitoring features.

By achieving these objectives, the proposed system aims to bridge the gap between technology and mental health support, providing students with a reliable tool for stress assessment and wellness improvement.

EXISTING SYSTEM

"Understanding the limitations of today is the first step toward building a better tomorrow." "Innovation begins where traditional solutions reach their boundaries."

In the current educational environment, student stress assessment is primarily performed through traditional and manual methods such as surveys, questionnaires, self-assessments, counseling sessions, and direct observation by teachers or mental health professionals. These approaches are widely used to identify students experiencing emotional or psychological challenges and to provide necessary support.

Most existing systems depend heavily on students voluntarily sharing their feelings and experiences. While counseling services and psychological assessments play an important role in mental health management, they often require significant time, effort, and professional intervention. In many cases, students may hesitate to discuss their stress levels openly due to fear, social stigma, lack of awareness, or privacy concerns. As a result, stress-related issues may remain unnoticed until they begin to negatively affect academic performance and personal well-being.

Additionally, traditional stress assessment methods generally focus on evaluating student conditions at a particular point in time rather than continuously monitoring changes in stress levels. This makes it difficult to identify early warning signs and implement preventive measures before stress becomes severe. Most existing approaches also lack automated analysis and personalized wellness recommendations that can guide students toward healthier lifestyle choices.

Furthermore, many conventional systems do not utilize modern technologies such as Artificial Intelligence and Machine Learning for predictive analysis. Without intelligent prediction mechanisms, institutions often rely on reactive approaches that address stress after it has already become a significant concern. This limits the effectiveness of stress management programs and reduces opportunities for early intervention.

The absence of an integrated platform that combines stress prediction, wellness recommendations, historical tracking, and user-friendly accessibility creates a gap in student wellness management. Therefore, there is a growing need for a smarter, data-driven, and proactive solution that can accurately assess stress levels and support students in maintaining their mental well-being.

Limitations of Existing System

1. Relies heavily on manual assessment and human intervention.
2. Limited capability for early stress detection and prevention.
3. Students may hesitate to disclose their mental health concerns.
4. Lack of continuous monitoring and historical tracking.
5. No automated prediction of stress levels.
6. Limited personalization in wellness guidance and recommendations.
7. Time-consuming and resource-intensive counseling processes.
8. Reactive rather than proactive approach to stress management.
9. Minimal utilization of Artificial Intelligence and Machine Learning technologies.
10. Difficulty in handling large-scale student wellness assessments efficiently.

These limitations highlight the necessity for an intelligent and automated system capable of predicting stress levels, providing personalized recommendations, and supporting proactive student wellness management.

PROPOSED SYSTEM

"The best solutions are those that identify problems before they become obstacles." "Artificial Intelligence empowers us to transform data into meaningful decisions and positive outcomes."

The proposed system, titled **"AI-Powered Student Stress Prediction and Wellness Recommendation System,"** is an intelligent web-based application designed to predict student stress levels using Machine Learning techniques and provide personalized wellness recommendations. The system aims to support early stress detection, encourage healthy lifestyle habits, and promote overall student well-being through a data-driven approach.

The application collects important lifestyle and academic parameters that significantly influence student mental health, including sleep hours, study hours, screen time, exercise hours, attendance percentage, social activity, and academic pressure. These inputs are processed by a trained Machine Learning model that analyzes the provided information and predicts the student's stress level.

Based on the prediction results, the system classifies students into one of four stress categories:

-  Low Stress
-  Moderate Stress
-  High Stress
-  Critical Stress

After identifying the stress category, the system generates personalized wellness recommendations tailored to the student's condition. These recommendations help students improve their daily routines, manage stress effectively, and adopt healthier lifestyle practices. Additionally, the system calculates a wellness score that provides a simple numerical representation of the student's overall wellness status.

The proposed solution is implemented using Python, Flask, HTML, CSS, SQLite, and Machine Learning technologies. The web-based architecture ensures easy accessibility, user-friendly interaction, and efficient processing of student data. To enhance usability, the system stores prediction records in a SQLite database, allowing users to review previous assessments and monitor stress trends over time through a prediction history dashboard.

Unlike traditional stress assessment approaches, the proposed system utilizes Artificial Intelligence to provide automated, accurate, and proactive stress evaluation. By identifying potential stress risks at an early stage, the system enables timely intervention and supports students in maintaining better mental health and academic performance.

Features of the Proposed System

1. AI-based student stress prediction using Machine Learning.
2. Classification into four stress categories.
3. Personalized wellness recommendations.
4. Wellness score calculation and display.
5. Interactive and user-friendly web interface.

6. Real-time prediction and analysis.
7. Prediction history storage using SQLite database.
8. Automated stress assessment without manual intervention.
9. Early identification of potential stress risks.
10. Scalable architecture for future enhancements and deployment.

Advantages of the Proposed System

1. Provides quick and accurate stress assessment.
2. Supports proactive stress management.
3. Reduces dependency on manual evaluation methods.
4. Encourages mental health awareness among students.
5. Maintains historical records for monitoring progress.
6. Delivers personalized wellness guidance.
7. Easy to use and accessible through a web application.
8. Cost-effective and scalable solution.
9. Integrates Artificial Intelligence for intelligent decision-making.
10. Contributes to improved student well-being and academic success.

The proposed system bridges the gap between technology and mental health support by providing an intelligent, efficient, and practical solution for student stress management. It demonstrates how Artificial Intelligence can be leveraged to create a healthier and more supportive educational environment.

METHODOLOGY

"A well-designed process transforms ideas into impactful solutions." "The power of Artificial Intelligence lies not only in prediction but also in enabling informed decision-making."

The methodology of the proposed "AI-Powered Student Stress Prediction and Wellness Recommendation System" focuses on collecting student-related data, analyzing it using Machine Learning techniques, predicting stress levels, and providing personalized wellness recommendations. The system follows a structured workflow that enables accurate stress assessment and supports proactive mental wellness management.

The overall methodology consists of data collection, data processing, Machine Learning-based prediction, recommendation generation, wellness score calculation, database storage, and result visualization. Each stage contributes to the efficient functioning of the system.

Step 1: Data Collection

The first stage involves collecting important lifestyle and academic parameters from students through a web-based input form. The system gathers the following information:

- Sleep Hours
- Study Hours
- Screen Time
- Exercise Hours
- Attendance Percentage
- Social Activity Hours
- Academic Pressure Level

These factors are considered significant indicators of student stress and overall well-being.

Step 2: Data Processing

After receiving the user inputs, the system converts the collected values into a structured numerical format suitable for Machine Learning analysis. The input data is validated and organized into a feature vector that can be processed by the trained prediction model.

This stage ensures consistency and accuracy before prediction is performed.

Step 3: Machine Learning-Based Stress Prediction

The processed input data is provided to the trained Machine Learning model. The model analyzes the relationship between the input features and previously learned stress patterns from the training dataset.

Based on the analysis, the model predicts one of the following stress categories:

-  Low Stress
-  Moderate Stress
-  High Stress
-  Critical Stress

This prediction serves as the foundation for further wellness assessment and recommendation generation.

Step 4: Wellness Recommendation Generation

After predicting the stress category, the system automatically generates personalized wellness recommendations. These recommendations are designed to encourage healthy lifestyle habits and stress management practices.

Examples include:

- Improving sleep quality and duration
- Maintaining regular physical activity
- Reducing excessive screen time
- Following a balanced study schedule
- Seeking support from mentors or counselors when necessary

The recommendations vary depending on the predicted stress level.

Step 5: Wellness Score Calculation

The system assigns a wellness score to represent the student's overall wellness condition.

Example:

- Low Stress → 95/100
- Moderate Stress → 75/100
- High Stress → 50/100
- Critical Stress → 25/100

The wellness score provides an easy-to-understand numerical representation of mental wellness status.

Step 6: Database Storage

The prediction results are stored in a SQLite database. The stored information includes:

- Stress Category
- Wellness Score
- Date and Time of Prediction

This enables the system to maintain historical records and allows users to track their stress assessments over time.

Step 7: Result Visualization

The final prediction results are displayed through an interactive dashboard. The dashboard presents:

- Stress Category
- Wellness Score
- Personalized Recommendations
- Stress-related Visual Indicators
- Prediction History

This user-friendly interface improves accessibility and enhances the overall user experience.

Methodology Workflow

Student Input Data ↓ Data Processing ↓ Machine Learning Model ↓ Stress Prediction ↓ Stress Category Classification ↓ Wellness Recommendation Generation ↓ Wellness Score Calculation ↓ Database Storage ↓ Dashboard Visualization

Summary

The proposed methodology combines Artificial Intelligence, Machine Learning, web technologies, and database management to create a comprehensive student wellness support system. By integrating prediction, recommendation, storage, and visualization into a single platform, the system enables early stress detection and promotes proactive mental health management among students.

TECHNOLOGY STACK

"Technology is most powerful when it works together to solve real-world challenges." "The right combination of tools transforms innovative ideas into impactful solutions."

The development of the "AI-Powered Student Stress Prediction and Wellness Recommendation System" requires the integration of multiple technologies to ensure accurate prediction, efficient data processing, user-friendly interaction, and reliable data storage. The selected technology stack combines Artificial Intelligence, web development frameworks, database management systems, and supporting libraries to create a complete end-to-end solution.

1. Python

Python serves as the primary programming language used for developing the application. It is widely recognized for its simplicity, readability, and extensive ecosystem of libraries that support Artificial Intelligence, Machine Learning, web development, and data processing.

Role in the Project: • Backend application development • Machine Learning implementation • Data processing and prediction • Database connectivity

Advantages: • Easy to learn and maintain • Rich Machine Learning ecosystem • Strong community support • High development productivity

2. Flask Framework

Flask is a lightweight Python web framework used to build the web application and connect the frontend with the Machine Learning model.

Role in the Project: • Handles HTTP requests and responses • Connects user inputs to the prediction model • Renders web pages dynamically • Manages application routing

Advantages: • Lightweight and flexible • Easy integration with Machine Learning models • Fast development process • Suitable for small and medium-scale applications

3. HTML (HyperText Markup Language)

HTML is used to design and structure the web pages of the application.

Role in the Project: • Creates user input forms • Displays prediction results • Presents recommendation information • Structures dashboard components

Advantages: • Simple and widely supported • Provides clear content organization • Easy integration with CSS and Flask templates

4. CSS (Cascading Style Sheets)

CSS is used to enhance the visual appearance of the web application and improve user experience.

Role in the Project: • Dashboard styling • Layout design • Responsive user interface • Visual presentation of results

Advantages: • Improves user engagement • Creates professional-looking interfaces • Enhances readability and usability

5. Machine Learning Model

The Machine Learning model is the core intelligence component of the system. It analyzes student-related parameters and predicts stress levels based on learned patterns from training data.

Role in the Project: • Stress level prediction • Pattern recognition • Data-driven decision making • Classification into stress categories

Advantages: • Automated analysis • High prediction efficiency • Supports early stress detection • Scalable for future improvements

6. Joblib

Joblib is used for saving and loading the trained Machine Learning model efficiently.

Role in the Project: • Stores trained model files • Loads model during application execution • Reduces retraining requirements

Advantages: • Fast model loading • Efficient serialization • Easy integration with Python applications

7. NumPy

NumPy is a scientific computing library used for numerical operations and data manipulation.

Role in the Project: • Input feature processing • Numerical computations • Array creation for model prediction

Advantages: • High-performance calculations • Efficient data handling • Essential for Machine Learning workflows

8. SQLite Database

SQLite is used as the database management system for storing prediction records and historical assessment data.

Role in the Project: • Stores stress predictions • Maintains wellness scores • Records prediction timestamps • Supports prediction history tracking

Advantages: • Lightweight and serverless • Easy setup and maintenance • Reliable local storage • Suitable for project-scale applications

Technology Stack Summary

Technology	Purpose
Python	Backend Development & Machine Learning
Flask	Web Application Framework
HTML	User Interface Structure
CSS	User Interface Design
Machine Learning Model	Stress Prediction
Joblib	Model Storage & Loading
NumPy	Data Processing
SQLite	Database Management

Summary

The selected technology stack provides a strong foundation for developing an intelligent, efficient, and scalable student wellness management system. The combination of Artificial Intelligence, web technologies, and database management enables the project to deliver accurate stress predictions, personalized recommendations, and a seamless user experience while maintaining simplicity and reliability.

IMPLEMENTATION

The implementation of the AI-Powered Student Stress Prediction and Wellness Recommendation System is carried out using a combination of Machine Learning, Flask web framework, and database integration to deliver a real-time intelligent application. **1. System Architecture**

The system follows a simple three-layer architecture: - Frontend Layer: HTML and CSS are used to design a user-friendly interface for collecting student input data. - Backend Layer: Flask (Python) processes user requests and handles communication between frontend and machine learning model. - Model Layer: A trained Machine Learning model (stress_model.pkl) is used to predict stress levels. **2. Machine Learning Model Implementation**

The model is trained using student behavioral and academic parameters such as sleep hours, study hours, screen time, exercise, attendance, and social activity. The trained model is saved using joblib and loaded into Flask for prediction. **3. Workflow of the System**

Step 1: User enters input data through the web form.

Step 2: Flask backend receives and preprocesses the data.

Step 3: The ML model predicts stress level (Low, Moderate, High, Critical).

Step 4: Based on prediction, wellness recommendations are generated.

Step 5: Results are displayed on the user interface.

4. Database Integration

SQLite database is used to store user inputs and prediction history. This allows users to track their stress patterns over time and analyze improvements. **5. Backend Implementation (Flask)**

Flask routes handle: - Home page rendering - Data input collection - Model prediction processing - Displaying results and recommendations **6. Output**

The system provides: - Predicted stress level - Wellness score - Personalized recommendations - History tracking capability **7. Tools Used**

Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, SQLite, Joblib. *"Implementation is where ideas become reality through code and logic."*

RESULTS AND DISCUSSION

The implemented AI-powered system successfully classifies student stress levels into multiple categories such as Low, Moderate, High, and Critical stress based on the input behavioral and academic parameters. The trained machine learning model demonstrates consistent performance in predicting stress levels with reliable accuracy during testing using sample datasets.

The Flask-based backend efficiently processes user inputs in real time and returns instant predictions, ensuring smooth interaction between the model and the user interface. The integration of the trained model (stress_model.pkl) with the web application allows seamless deployment, making the system practical for real-world usage in academic environments.

Key Results Observed

- The model successfully differentiates between varying stress levels with clear classification boundaries.
- Real-time prediction output is generated within seconds, ensuring fast response.
- The system provides personalized wellness recommendations based on predicted stress levels.
- The user interface effectively collects input data and displays meaningful results in a structured format.

Discussion

The results indicate that machine learning can be effectively applied in the domain of mental health monitoring, especially for early stress detection among students. The system not only predicts stress levels but also bridges the gap between detection and intervention by suggesting wellness activities such as mindfulness exercises, study planning, rest routines, and lifestyle adjustments.

However, the accuracy of predictions depends heavily on the quality and diversity of training data. With more real-world datasets, the model can be further improved for higher precision and better generalization. Additionally, integrating deep learning techniques or continuous user feedback could enhance future versions of the system.

This project highlights the potential of AI in educational wellness systems, where technology acts as a supportive tool rather than a replacement for human guidance.

"Technology becomes powerful when it understands human emotions."

"Early detection is the first step toward better mental well-being."

FUTURE SCOPE

The AI-Powered Student Stress Prediction and Wellness Recommendation System has strong potential for further enhancement and real-world scalability. Future improvements can make the system more accurate, intelligent, and widely applicable across different educational and professional environments.

1. Integration of Real-Time Data

Future versions can collect real-time behavioral data such as app usage patterns, sleep tracking, and academic performance logs to improve prediction accuracy and make the system more dynamic.

2. Advanced AI and Deep Learning Models

The current machine learning model can be upgraded to deep learning architectures (such as neural networks) to handle complex patterns in student behavior and improve stress classification precision.

3. Mobile Application Development

Developing a mobile app version (Android/iOS) will make the system more accessible to students, allowing continuous monitoring and instant wellness suggestions anytime, anywhere.

4. Personalized Mental Health Chatbot

A conversational AI chatbot can be integrated to provide emotional support, daily check-ins, motivational messages, and guided relaxation techniques based on user stress levels.

5. Integration with Wearable Devices

Wearable devices like smartwatches can be connected to track physiological signals such as heart rate, sleep quality, and activity levels for more accurate stress detection.

6. Institutional Dashboard for Teachers & Counselors

A secure dashboard can be developed for academic institutions to monitor overall student well-being trends while maintaining privacy and ethical data handling.

7. Multilingual Support

To increase accessibility, the system can be extended to support multiple regional and global languages, enabling wider adoption among students.

8. Predictive Intervention System

Instead of only predicting stress, future systems can proactively suggest preventive actions before stress reaches a critical level.

"The future of mental wellness lies in intelligent systems that not only detect stress but also prevent it."

"AI is not just about prediction, it is about protection and prevention."

CONCLUSION

The AI-Powered Student Stress Prediction and Wellness Recommendation System successfully demonstrates how machine learning can be applied to address mental health challenges in academic environments. The system effectively predicts student stress levels based on input parameters and provides appropriate wellness recommendations to support emotional and academic well-being.

By integrating a trained machine learning model with a Flask-based web application, the project achieves real-time prediction with a user-friendly interface. This ensures that students can easily understand their stress condition and receive immediate guidance for improvement.

The system highlights the importance of early stress detection, which can help reduce academic pressure, improve productivity, and promote a healthier learning environment. It also shows how AI can act as a supportive tool in mental health awareness rather than replacing professional counseling.

Overall, this project proves that intelligent systems can play a significant role in identifying stress patterns and encouraging proactive wellness management among students.

"A healthy mind is the foundation of successful learning."

"When technology understands well-being, education becomes more human-centered."

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